# Opgaver Kapitel 9

Klassisk Fysik

41. Find the center of mass of a pentagon with five equal sides of length $a$, but with one triangle missing (Fig. 9.24). (*Hint:* See Example 9.3, and treat the pentagon as a group of triangles.)

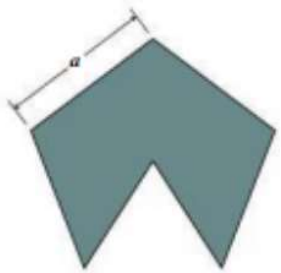


**FIGURE 9.24** Problem 41

88. In a ballistic pendulum demonstration gone bad, a 0.52-g pellet, fired horizontally with kinetic energy 3.25 J, passes straight through a 400-g Styrofoam pendulum block. If the pendulum rises a maximum height of 0.52 mm, how much kinetic energy did the pellet have after emerging from the Styrofoam?

50. On an icy road, a 1200-kg car moving at 50 km/h strikes a 4400-kg truck moving in the same direction at 35 km/h. The pair is soon hit from behind by a 1500-kg car speeding at 65 km/h, and all three vehicles stick together. Find the speed of the wreckage.

72. A proton (mass 1 u) moving at 6.90 Mm/s collides elastically head-on with a second particle moving in the opposite direction at 2.80 Mm/s. After the collision, the proton is moving opposite its initial direction at 8.62 Mm/s. Find the mass and final velocity of the second particle.

92. Model rocket motors are specified by giving the impulse they
DATA provide, in N·s, over the entire time the rocket is firing. The table below shows the results of rocket-motor tests with different motors used to launch rockets of different masses. Determine two data-based quantities that, when plotted against each other, should give a straight line and whose slope should allow you to determine $g$. Plot the data, establish a best-fit line, and determine $g$. Assume that the maximum height is much greater than the distance over which the rocket motor is firing, so you can neglect the latter. You're also neglecting air resistance—but explain how that affects your experimentally determined value for $g$.

| Impulse, $J$ (N · s) | 4.5 | 7.8 | 4.5 | 7.8 | 11 |
|---|---|---|---|---|---|
| Rocket mass (g) (including motor) | 180 | 485 | 234 | 234 | 485 |
| Maximum height achieved (m) | 22 | 13 | 19 | 51 | 23 |